

Prove Me Wrong

A Framework for the Next Paradigm in Computing and the Nature of Reality

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Date: March 30, 2026

Version: 1.0

Abstract

Three valued logic is a more accurate model of reality than binary logic. Binary was built by observers embedded in a 5th dimensional slice of a 7+1 dimensional universe, unable to account for spin, entropy drag, or the frame drag between the opposing arrows of time and thermodynamics. This paper presents a complete theoretical framework for a ternary computing architecture grounded in that dimensional structure, including a spherical logic gate implementing solid state probability collapse, a triarity parity mechanism for resolving indeterminate states, and a Clifford algebraic mapping of the full dimensional stack via Bott periodicity. The framework demonstrates that the Maybe state -- long treated as a computational liability -- is where reality itself performs its work. Moore's limit is not an engineering problem. It is a symptom of a logic system that was never adequate to describe the universe it runs inside.

Keywords: three-valued logic, ternary computing, Bott periodicity, Clifford algebra, probability collapse, dimensional ontology, Higgs dimension, entropy, frame dragging, triarity parity, logic sphere, Moore's limit

I. The Problem With Yes and No

Our entire world is based on the same thing it was in the Paleolithic era. Who has the biggest rock wins.

After millennia of civilization, some guys in a laboratory discovered solid state electronics. Forty years later, some other guys made the first integrated circuit.

They just produced 5nm chips. It is March 2026.

They have reached the atomic limit of what silicon can do. They have reached Moore's limit. Anything short of a paradigm shift in how we approach computing is finished.

First: Stop thinking logic has to be Yes or No. That itself is a logical fallacy. Multi-valued logic is more apt for reality. Yes, No, and Maybe is how reality actually works.

Second: Maybe is the only infinite state. It is infinite -- yet definable within a ternary machine.

Third: Maybe resolves into two directional states -- Maybe-Yes and Maybe-No -- mathematically defined by .3333... and .6666... Since reality itself dictates that .9999... equals 1, the math holds.

II. The Observer Problem

Before the logic, before the gate, before the architecture -- there is a more fundamental problem.

Binary logic was built by observers who never accounted for their own observational position.

Reality as we experience it is a 5th dimensional slice -- the Higgs dimension, where mass is assigned and physical existence is constructed. We are observers embedded inside that slice. We built our logic from inside it, looking outward, and assumed that what was visible from that position was all there was to define.

From inside the 5D slice, Yes and No look complete. From inside the slice there is no observable spin. There is no entropy drag pulling in the opposite direction from outside the observational frame. There is no gravity eternally drawing everything toward the reset. None of that is visible to a 2D logic system built by a 5D observer.

So they built a logic that is technically correct within the slice and catastrophically incomplete as a description of reality.

A 2D logic cannot account for the full 3D space that the 5D slice sits inside. You are trying to describe a volume with a flat plane. The missing dimension is exactly where spin lives. Where entropy drag lives. Where the irreversibility of collapse originates.

The Maybe state is not exotic or undefined from outside the slice. From outside the slice Maybe is the most natural state imaginable. It is what spin looks like when you cannot see the spin.

Binary logic does not describe reality. It describes the limitations of the observer who invented it -- and those limitations got mistaken for a fundamental truth about the universe.

We only get a 5D slice of reality. Everything we have built computationally reflects that constraint. The paradigm shift is not just a new gate architecture. It is a new observational framework that accounts for the full dimensional stack the slice sits inside.

III. The Logic Sphere

A sphere is a miniature version of reality. Gravity and the Higgs field make the sphere the natural shape of everything in the universe. The logic gate mirrors the substrate it runs on. This is not engineering convenience. It is ontological consistency.

Four points within that sphere define the gate -- three inputs, one output. A simplex locates those four points mathematically. The simplex is virtual. It is a descriptor, not a physical object inside the gate. Whether the construction requires a regular simplex or whether Heronian simplices also satisfy it remains an open question.

The gate operates in balanced ternary. Inputs are -1, 0, or 1. Three input points on the sphere carry their values into the bulk material. The output point resolves what the three inputs produce together -- one of approximately 20,000 possible states in a three-valued logic system.

That state space partitions into three equal thirds by the nature of 3VL itself:

6,666 states resolve to defined Yes. They are locked.

6,666 states resolve to defined No. They are locked.

6,666 states are undefined Maybe. This third splits in half -- 3,333 states are Maybe-Yes, undefined but leaning toward 1, and 3,333 states are Maybe-No, undefined but leaning toward -1.

The defined states need no resolution. Triarity parity exists solely to resolve the 6,666 undefined states.

Three logic spheres run in series. Boom, boom, boom. Each sphere collapses its input state independently. Majority rule decides the output. Two Maybe-Yes votes and one Maybe-No resolves to Maybe-Yes. Two Maybe-No votes and one Maybe-Yes resolves to Maybe-No. The indeterminate state does not stay indeterminate. It gets outvoted across three sequential collapse events.

This is solid state probability collapse. Not quantum. Not probabilistic in the statistical sense. Deterministic resolution of a genuinely indeterminate state through classical geometry and majority logic.

Using Kleene and Priest logics to bound the undefined state space, the useful logical outputs reduce to approximately 6,666 -- the Maybe partition -- which is precisely what the triarity mechanism is built to resolve.

IV. Why Collapse Is Irreversible

The collapse is irreversible. Not because of engineering constraints. Because of the structure of reality itself.

The universe operates under two immutable laws pulling in opposite directions. Time flows in one direction. Entropy flows in the opposite direction. Reality as we experience it is the construction that exists in the tension between these two opposing flows.

Both flows are moving. Both exert force on everything caught between them. That differential -- two infinite opposing forces acting on the same medium -- imparts spin to everything in reality. This is not metaphor. Spin is not a primitive property of matter. Spin is what happens to matter and energy when caught between the opposing flows of time and entropy. Spin is the physical signature of being real.

When probability collapses inside the three-sphere series, it locks into that spin. The collapse acquires angular momentum from the frame drag between time and entropy at the moment of resolution. You cannot uncollapse it for the same reason you cannot unspin a gyroscope without applying external work. The angular momentum is conserved and directional.

Simultaneously, gravity at dimension 7 is eternally drawing every collapsed state back toward the 1D and 2D vector space. This draw is continuous. It does not begin at the event horizon. It is happening at all times to every collapsed state from the moment it locks.

Collapse is irreversible because spin is irreversible without external work -- and because gravity is already pulling every locked state toward the reset.

Standard quantum mechanics asserts collapse is irreversible without explaining why. Decoherence theory describes the direction without explaining the mechanism. This

framework explains both. The mechanism is frame drag. The direction is determined by the spin imparted at the moment of collapse. Gravity ensures there is no path back.

V. The Dimensional Stack and Bott Periodicity

The opposing flows of time and entropy are not merely forces. They occupy distinct dimensions. The full structure of reality maps across seven dimensions plus one -- grounded in Bott periodicity, the mathematical principle governing the periodic repetition of Clifford algebras $Cl(p,q)(R)$ across eight positions.

The clock runs with $R+R$ at the top. Not because it is the end but because it is the reset point -- the origin from which every cycle flows and to which every cycle returns. Reading clockwise:

$R+R$ -- 7+1. The Event Horizon.

Direct sum of two independent real number systems. Two complete copies of R existing simultaneously without mixing -- the collapsing cycle and the expanding cycle occupying the same position for one abrupt moment before the fold snaps. This is not a gradual transition. Gravity has been drawing continuously from dimension 7 and then it completes. Everything folds back through the vector space and the new cycle begins. What astrophysics calls Hawking radiation is the re-expansion signal of that new cycle starting. The only point in the entire dimensional stack where simultaneous existence breaks down.

R -- 1D and 2D. Vector Energetics.

Real numbers. The number line and the plane share the same algebraic designation because they are the same family -- real numbers bounded in one and two dimensions. Length, width, depth. The output layer. The projection screen onto which everything above renders as physical reality. The stage, not the machinery.

C -- 3D.

Complex numbers. The complex plane introduces a component perpendicular to the real number line -- an axis that cannot be fully observed from within R alone. 3D space is where complex numbers live. Commutative.

H -- 4D. Time.

Quaternions. Non-commutative. AB does not equal BA . The arrow of time is not a law imposed from outside. It is algebraically mandated by the quaternionic structure at this position. Order of operations is irreversible because the algebra makes it so. Traversed in the +1p direction clockwise.

H+H -- 5D. The Higgs Dimension.

Split quaternionic. Two full quaternionic systems in tension simultaneously. This is reality as construction -- the interference layer between the non-commutativity of Time pulling from H at 4D and the non-commutativity of Entropy pulling from H at 6D. Mass is not a property of things in spacetime. It is the interference pattern produced by two opposing non-commutative structures acting across this position at once. We are observers embedded inside this slice. Everything we have ever measured, built, or experienced exists here.

H -- 6D. Entropy.

Quaternions again. Non-commutative. AB does not equal BA . The critical fact: this position and Time at 4D are both H -- both quaternionic, both non-commutative -- but they are reached by traversing the Bott clock in opposite directions. +1p clockwise reaches Time at 4D. +1q counter-clockwise reaches Entropy at 6D. Same algebra. Opposite orientation. The opposing arrows of time and entropy are not asserted. They are written directly into the mathematics of Clifford algebra. Energy is timeless. It does not flow through time. It flows against it dimensionally because the non-commutativity runs the other way around the clock.

C -- 7D. Gravity.

Complex numbers again. Not a force. Not spacetime curvature. A dimension. Complex numbers have a real component and an imaginary component perpendicular to the real axis. We observe the real component of gravity -- mass attracts mass, things fall, orbits curve. The imaginary component is perpendicular to our observational position inside the 5D Higgs slice. We cannot see it from here. We only feel its shadow. This is why gravity appears the weakest force at human scales -- we are measuring only the real part of a complex dimensional structure. The imaginary part is doing the actual work: the eternal draw of everything back toward R+R at the top of the clock. Gravity is one position away from the reset at all times. That proximity is not incidental. It is the mechanism of the draw.

The two C positions -- 3D and 7D -- share the same algebraic structure and are both partially invisible from inside the 5D slice. Physical space and gravity are both complex

dimensional structures. Both have components perpendicular to what a 5D observer can directly measure. That symmetry is structural.

All dimensions exist simultaneously -- except at the R+R fold at the top of the clock.

VI. The Gate and the Universe Are Doing the Same Thing

The three-sphere series resolving Maybe states through sequential collapse is not analogous to how the universe resolves indeterminacy under thermodynamic pressure. It is the same operation at a different scale.

A single sphere tells you Maybe. It cannot tell you which way Maybe is leaning.

Three spheres in series, each receiving the Maybe output of the previous, with time and entropy acting between each collapse event -- the pattern of how Maybe propagates through the series encodes the thermodynamic arrow. The accumulated spin across three sequential collapse events points toward the more probable resolution.

Maybe-Yes at .666... is the higher entropy state. More microstates available. The universe preferentially collapses toward it for the same reason heat flows from hot to cold -- not because of a rule imposed from outside but because there are more ways to be that than to be the alternative.

Maybe-No at .333... is the lower entropy state. Rarer. More ordered. It requires work to reach and work to maintain.

The logic gate is a physical model of how indeterminacy resolves under thermodynamic pressure. The cosmological framework and the computing architecture are not linked by analogy. They are the same mathematics operating at different scales.

This is the paradigm shift. Not ternary arithmetic. Not a faster processor. The locus of computation moves from the defined states to the resolution of the undefined one. Binary logic is engineered to avoid the middle ground. This architecture makes the middle ground the location where the actual computation happens.

Maybe is not a bug. Maybe is where reality does its work.

VII. What Remains Open

The material substrate of the logic sphere is an open engineering constraint. The framework defines what the material must do -- accept field input at three tetrahedral points, propagate and combine those fields through its bulk volume, produce a deterministic output state at the fourth point, and support three stable resolution states. What material satisfies those constraints is a problem that falls out of the framework naturally once the framework is complete.

Whether the simplex locating the four gate points must be regular or whether Heronian simplices also satisfy the construction mathematically is unresolved.

The precise algebraic mapping of the approximately 20,000 state space onto the sphere and the formal proof of the triarity parity resolution mechanism under Kleene and Priest logics requires full formal treatment.

These are open questions. They are stated as such because intellectual honesty demands it and because defining the boundary of what is solved is as important as the solution itself.

The framework is complete. The engineering follows.

Prove me wrong.